Day 1-5: Build and Configure a small Office Network Deliverable

Q1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Ans:

Step 1: Create the Network Topology

- Place 1 Switch and 5 PCs (PC0, PC1, PC2, PC3, and PC4) in Cisco Packet Tracer.
- Connect all PCs to the Switch using Copper Straight-Through cables.

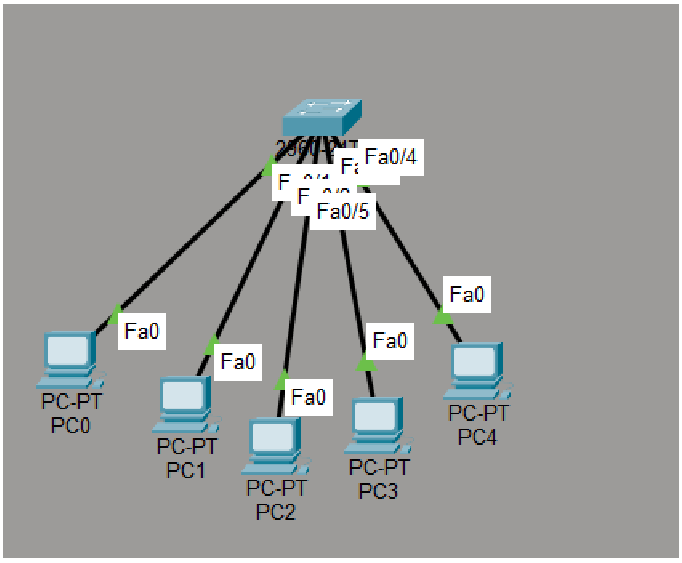


Figure 1: Network Topology

Step 2: Configure Static IP Addresses

Open each PC and navigate to:

Desktop → IP Configuration → Static

Assign the following IP addresses:

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.11	255.255.255.0	192.168.1.1
PC2	192.168.1.12	255.255.255.0	192.168.1.1
PC3	192.168.1.13	255.255.255.0	192.168.1.1
PC4	192.168.1.14	255.255.255.0	192.168.1.1

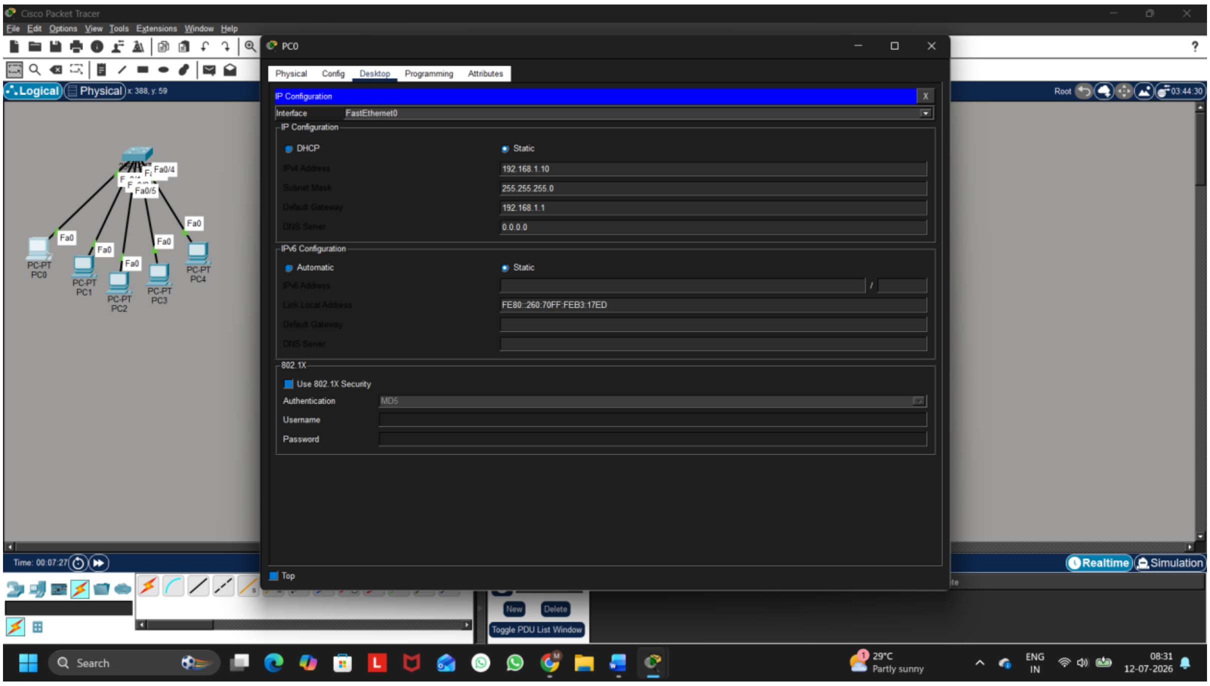


Figure 2: Static IP Configuration

Step 3: Verify Network Connectivity

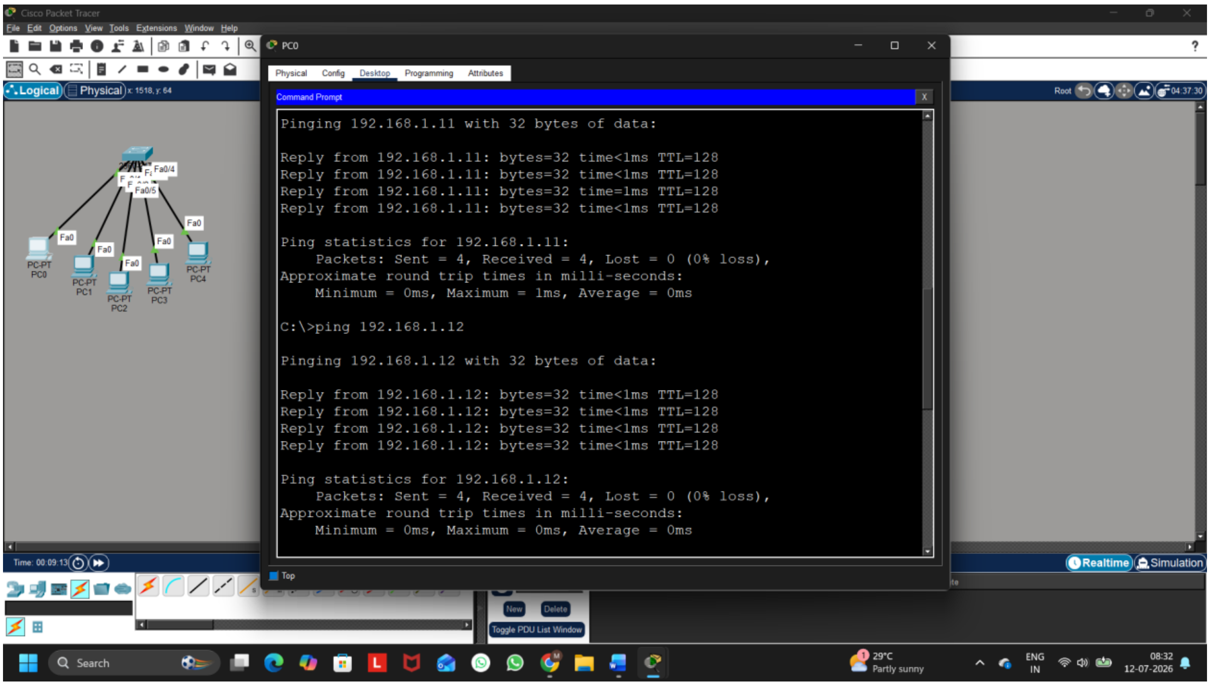


Figure 3: Successful Ping Verification for PC1 and PC2

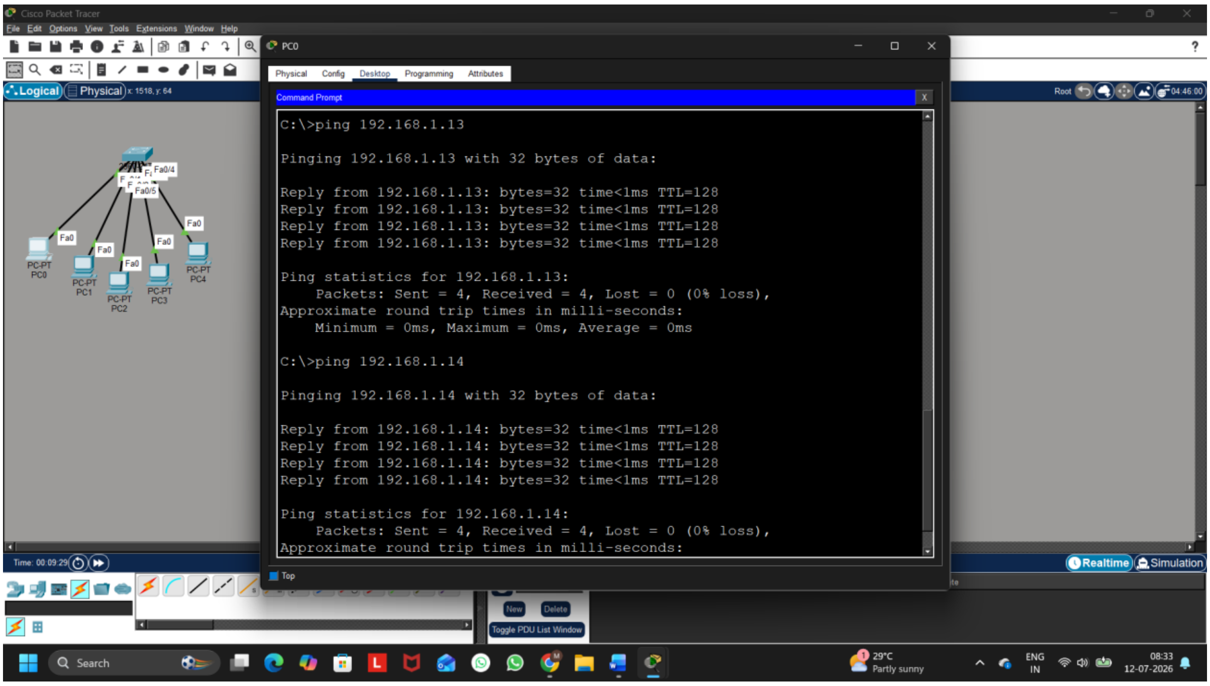


Figure 3: Successful Ping Verification for PC3 and PC4

Five devices were successfully assigned static IP addresses within the same subnet. Connectivity between the devices was verified using the ping command, and all devices responded successfully, confirming proper network communication.

Q2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Ans:

Step 1: Create the Network Topology

- Place the following devices in Cisco Packet Tracer:
  - 1 Server
  - 1 Switch
  - 3 PCs (PC5, PC6, and PC7)
- Connect all devices to the Switch using Copper Straight-Through cables.

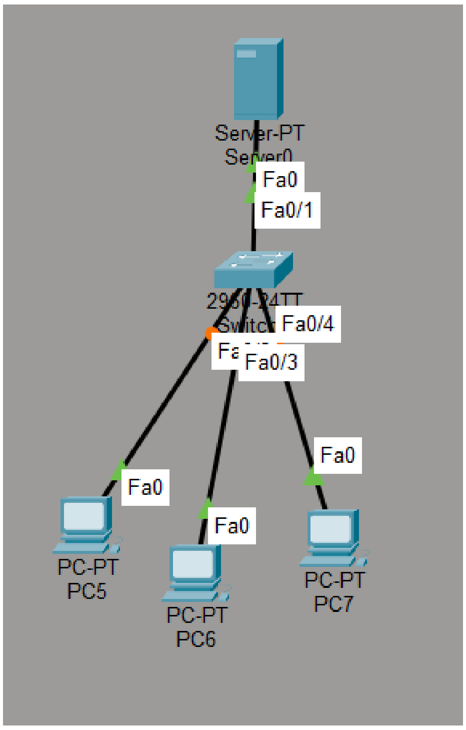


Figure 1: Network Topology

Step 2: Configure the Server

Open Server → Desktop → IP Configuration and configure:

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

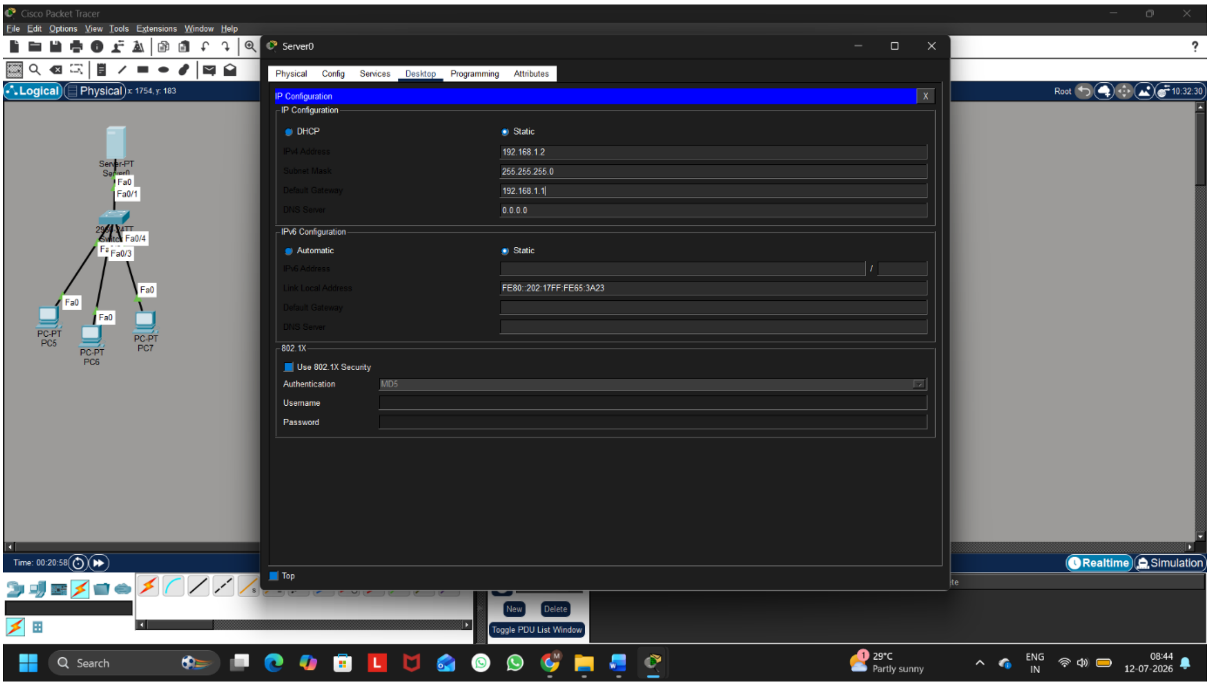


Figure 2: Server IP Configuration

Step 3: Configure the DHCP Server

Go to:

Server → Services → DHCP

Turn **DHCP Service = ON**

Create a DHCP Pool with the following configuration:

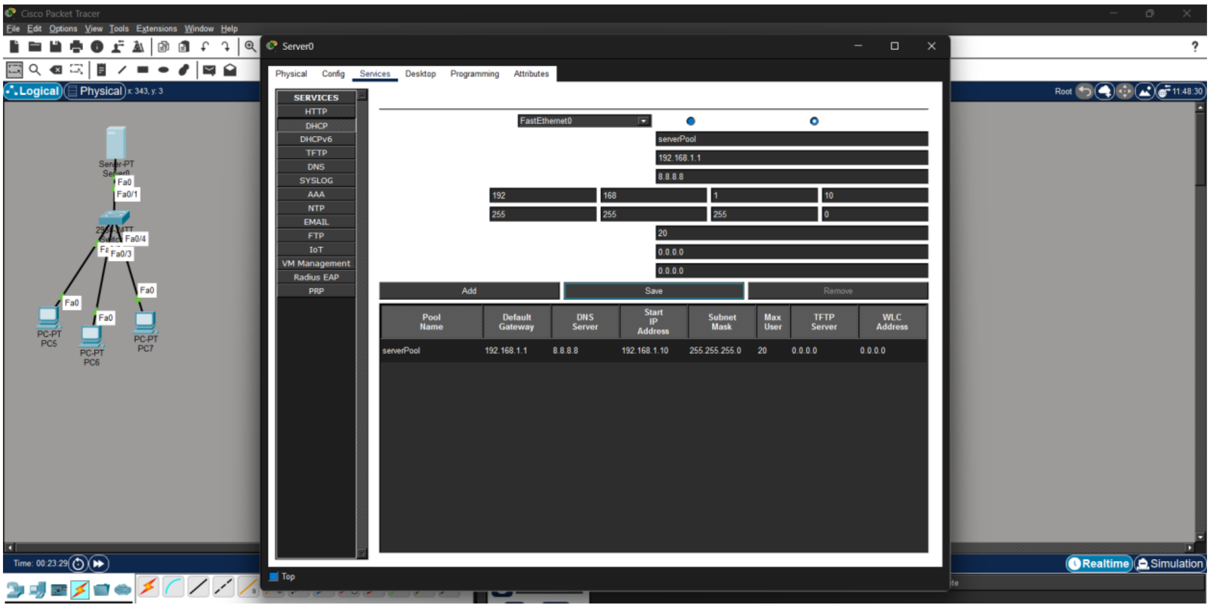


Figure 3: DHCP Server Configuration

Step 4: Configure the Client PCs

For each PC:

Desktop → IP Configuration → DHCP

The DHCP server will automatically assign an IP address.

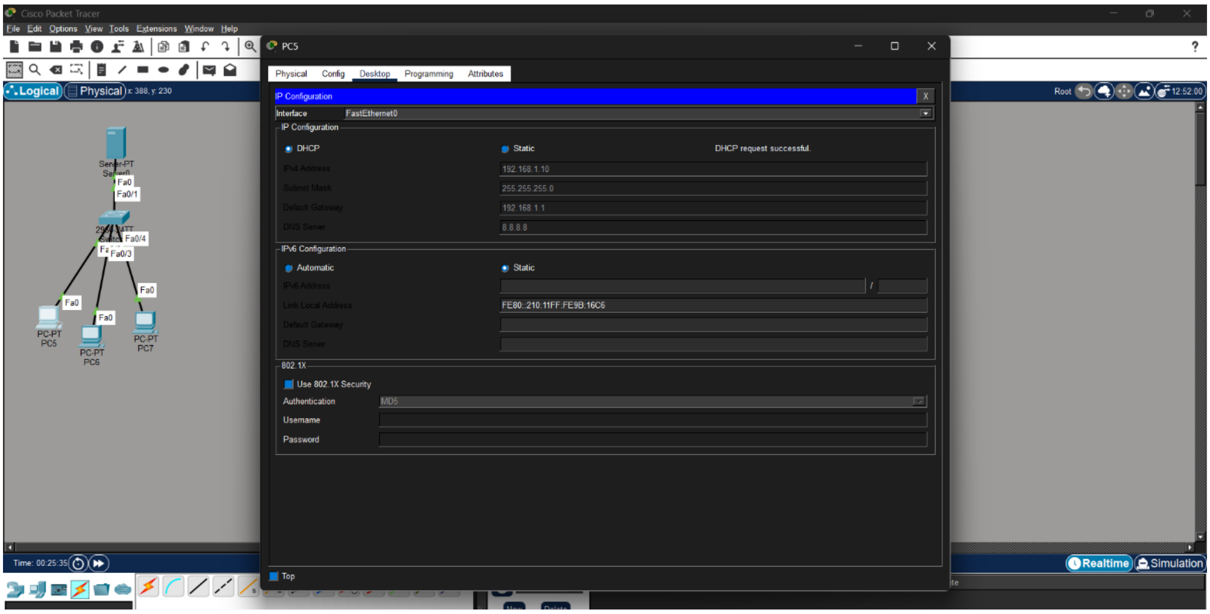


Figure 4: PC5 Receiving DHCP Address

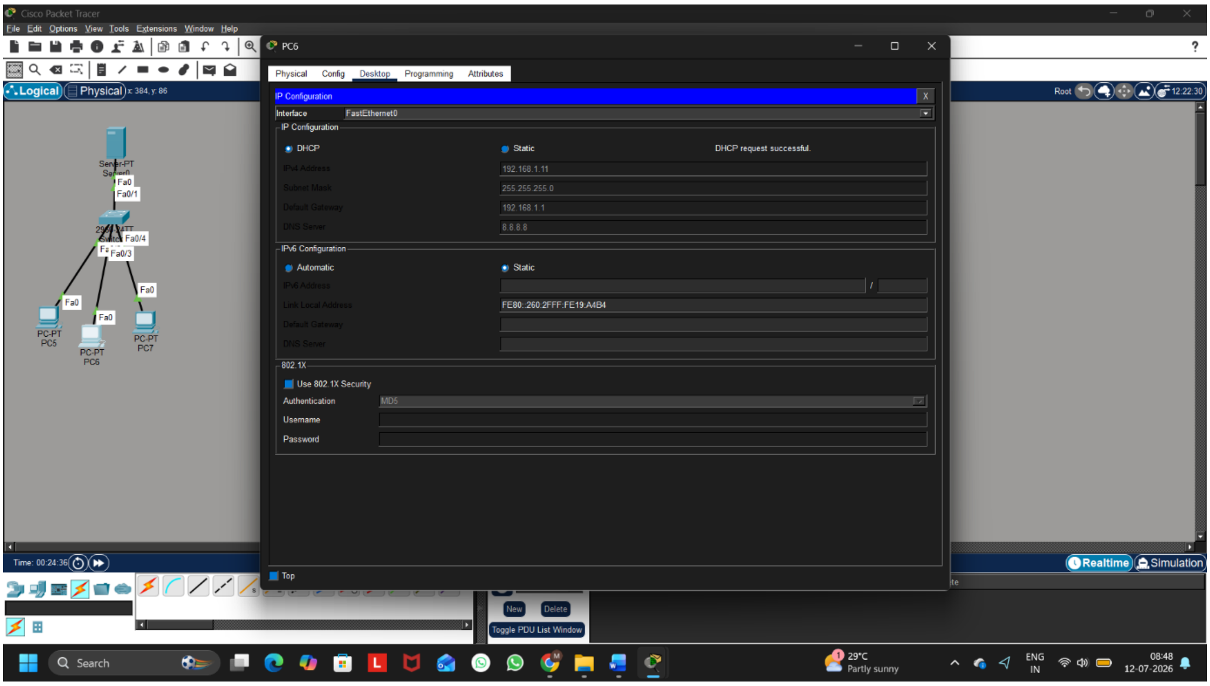


Figure 5: PC6 Receiving DHCP Address

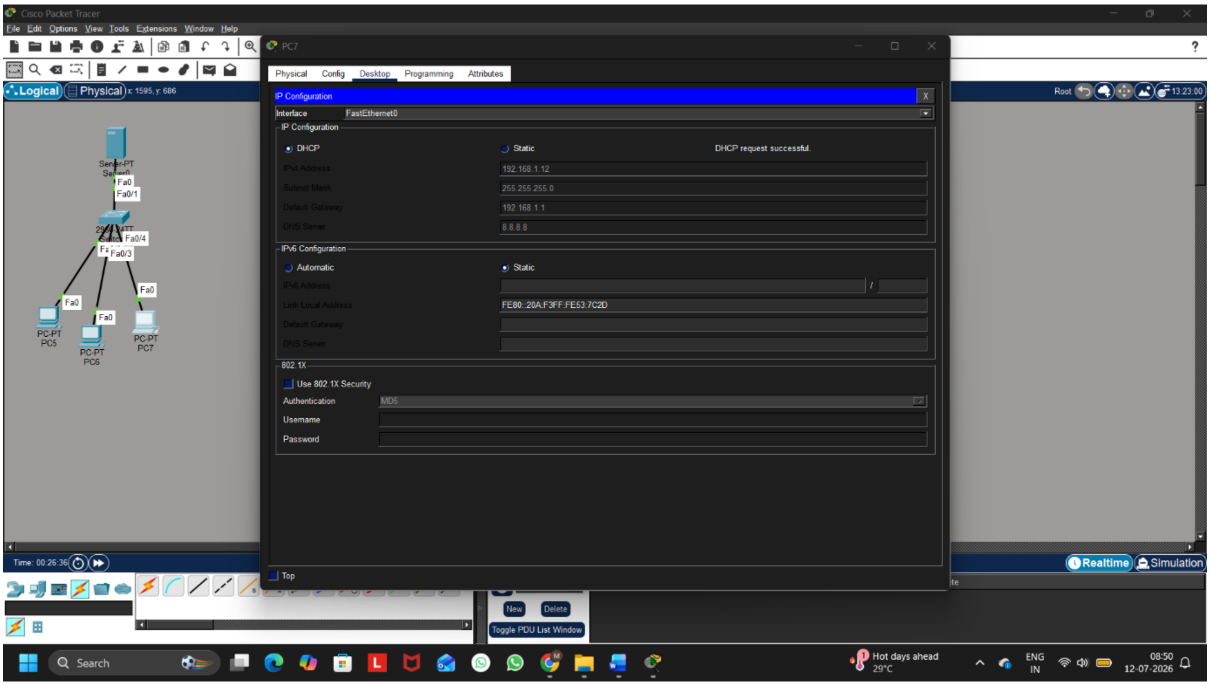


Figure 6: PC7 Receiving DHCP Address

Device	Assigned IP Address
PC	192.168.1.10
PC6	192.168.1.11
PC7	192.168.1.12

**Q3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.**

Ans:

Step 1: Create the Network Topology

Add the following devices:

- 1 Server
- 1 Switch
- 1 PC

Connect all devices using Copper Straight-Through cables.

Step 2: Configure the Server

Go to

Desktop → IP Configuration

Configure:

Setting	Value
IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

Step 3: Configure the DNS Service

Go to

Services → DNS

Turn

DNS = ON

Add the following DNS Record:

Name	Address
musicplayer.local	192.168.1.2

Click

Add

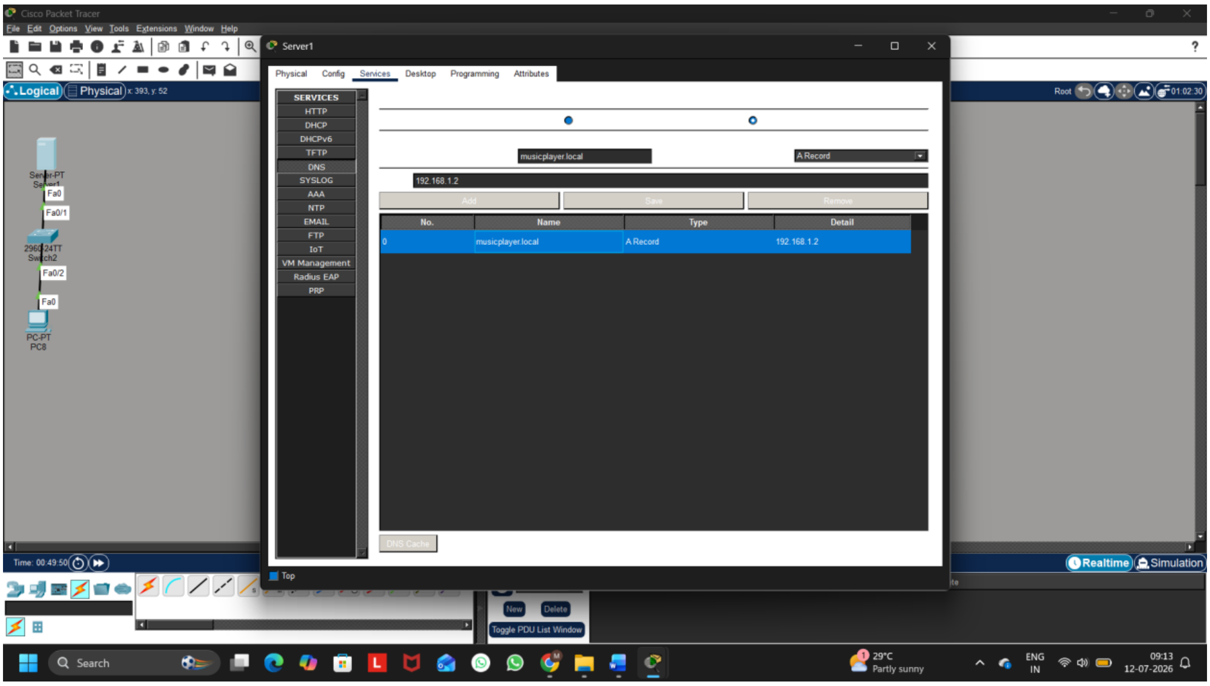


Figure 1: DNS Service Configuration

Step 4: Configure the HTTP Service

Go to

Services → HTTP

Turn

HTTP = ON

(Optional)

Edit index.html

Example

<h1>Welcome to Music Player</h1>

Save the page.

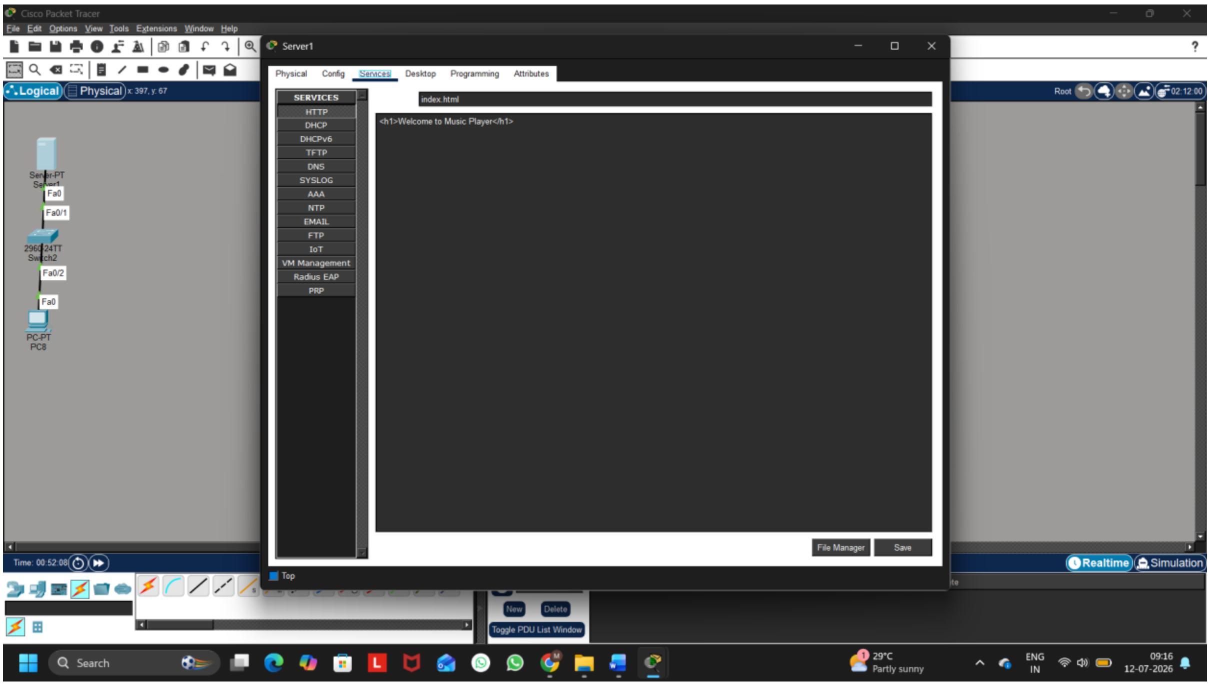


Figure 2: HTTP Service Configuration

Step 5: Configure the Client PC

Desktop

IP Configuration

Static

Configure

Setting	Value
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.2

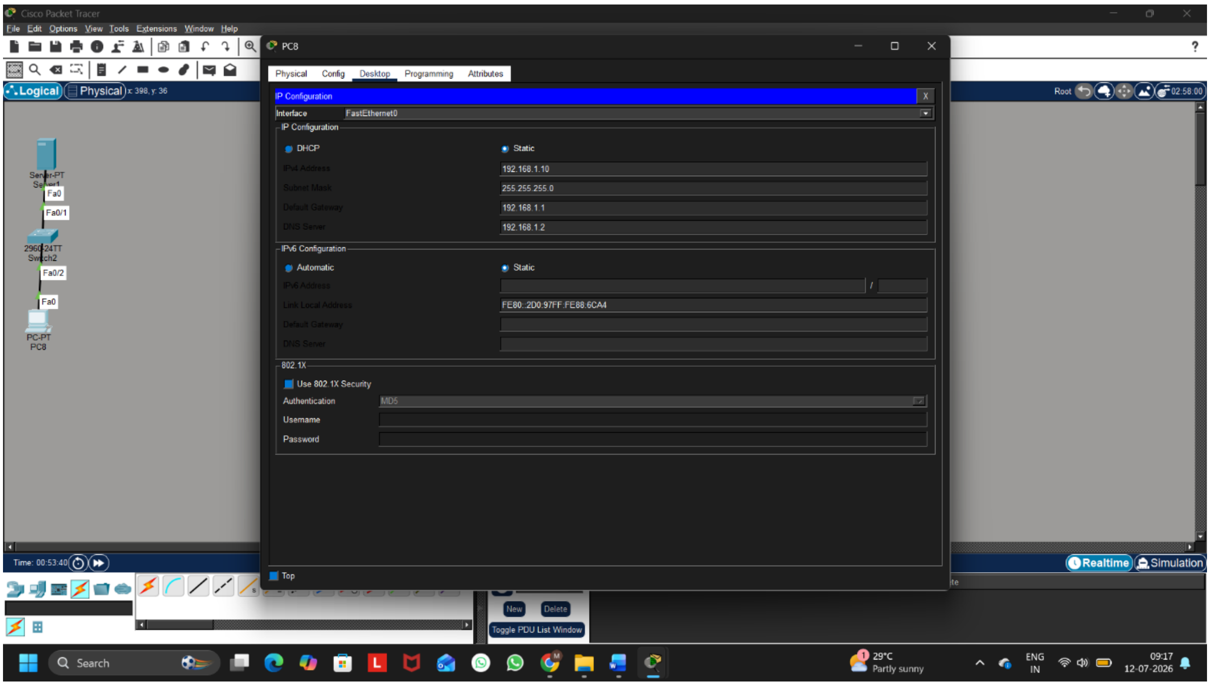


Figure 3: Client IP Configuration

Step 6: Verify DNS Resolution

Desktop

Command Prompt

Run

nslookup musicplayer.local

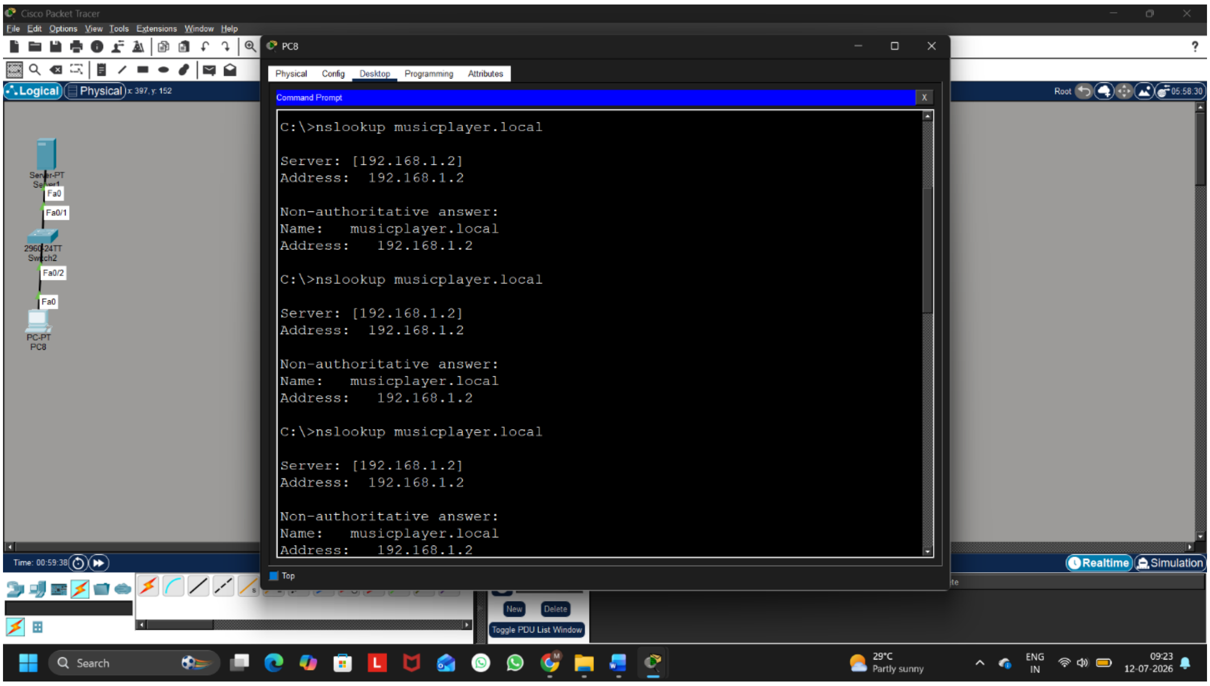


Figure 4: Successful DNS Lookup

Step 7: Open the Website

Desktop

Web Browser

Type

http://musicplayer.local

The browser opens the local webpage hosted on the Server.

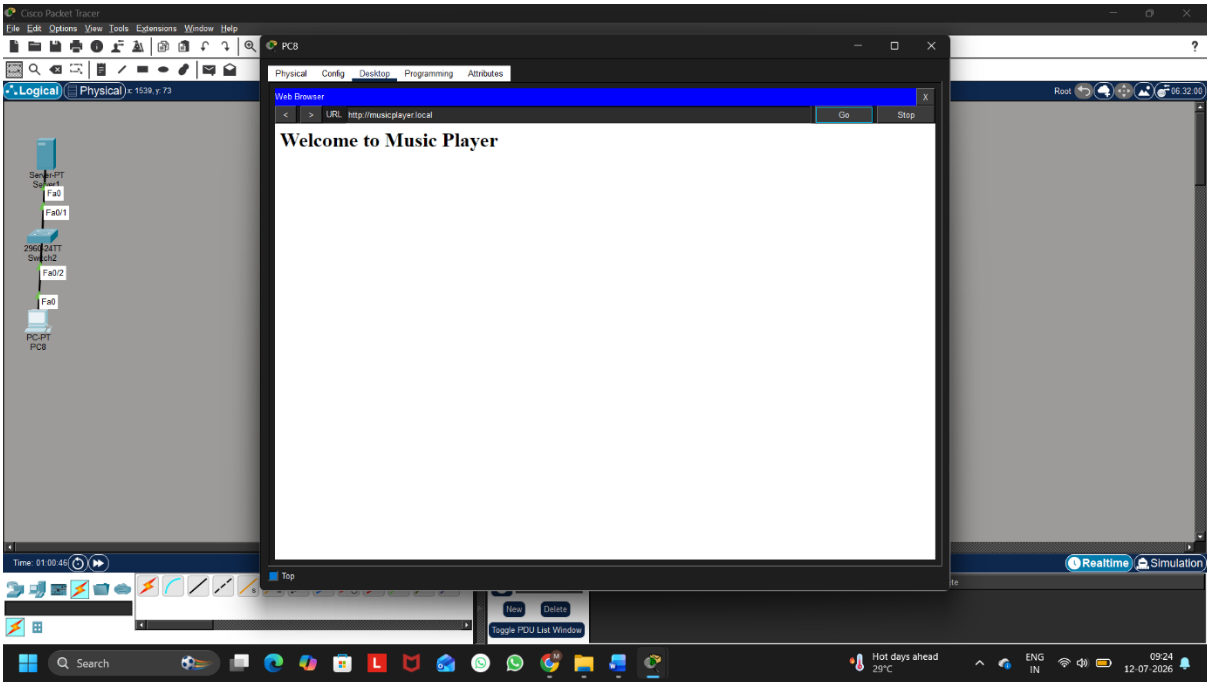


Figure 5: Local Web Page Opened Using Custom Hostname

The DNS service was successfully configured in Cisco Packet Tracer. A custom hostname (musicplayer.local) was mapped to the local server IP address (192.168.1.2). The hostname was verified using the nslookup command, and the webpage was successfully accessed through the browser using the custom hostname.

**Q4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

Hint: Use ping or file sharing to test connectivity between devices in different VLANs.**

Ans:

Step 1: Create the Network Topology

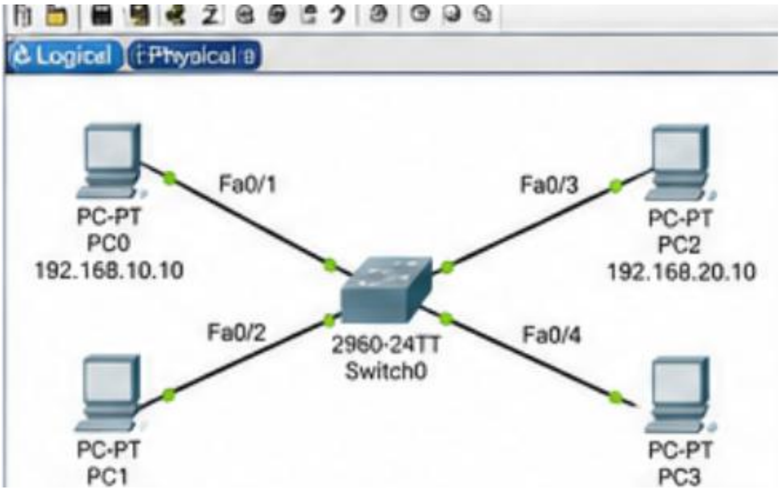
Add:

- 1 Switch (2960)
- 4 PCs

Connections:

- PC0 → Fa0/1
- PC1 → Fa0/2

- PC2 → Fa0/3
- PC3 → Fa0/4



Step 2: Configure Static IP Addresses
Assign the following IP addresses.
VLAN 10 (Workstations)

Device	IP Address	Subnet Mask
PC0	192.168.10.10	255.255.255.0
PC1	192.168.10.11	255.255.255.0

VLAN 20 (Smart Devices)

Device	IP Address	Subnet Mask
PC2	192.168.20.10	255.255.255.0
PC3	192.168.20.11	255.255.255.0

Step 3: Create VLANs
Click the Switch
CLI
Type:
enable

configure terminal

vlan 10
name Workstations

exit

vlan 20
name SmartDevices

exit

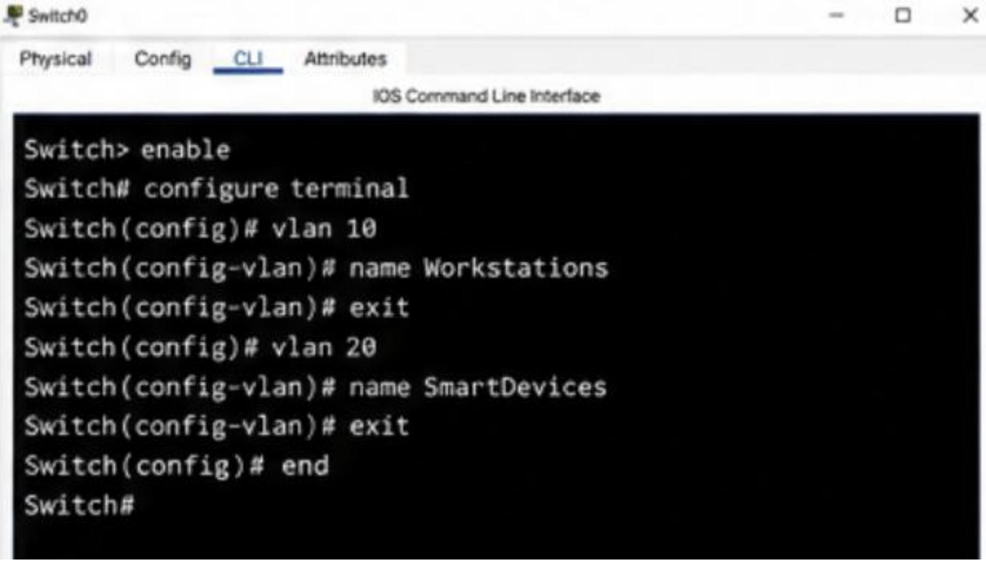


Figure 3: VLAN Creation

Step 4: Assign Ports to VLANs

Type:

interface fa0/1
switchport mode access
switchport access vlan 10
exit

interface fa0/2
switchport mode access
switchport access vlan 10
exit

interface fa0/3
switchport mode access
switchport access vlan 20
exit

interface fa0/4
switchport mode access
switchport access vlan 20
exit

```
Switch# configure terminal
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# interface fa0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# end
Switch#
```

Step 5: Verify VLAN Configuration

Type:

show vlan brief

Output should show:

VLAN 10

Fa0/1

Fa0/2

VLAN 20

Fa0/3

Fa0/4

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch# show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   Workstations                     active    Fa0/1, Fa0/2
20   SmartDevices                     active    Fa0/3, Fa0/4
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
Switch#
```

Figure 5: show vlan brief Output

Step 6: Test Communication Within the Same VLAN

From PC0

ping 192.168.10.11
Reply should be successful.
From PC2
ping 192.168.20.11
Reply should also be successful.

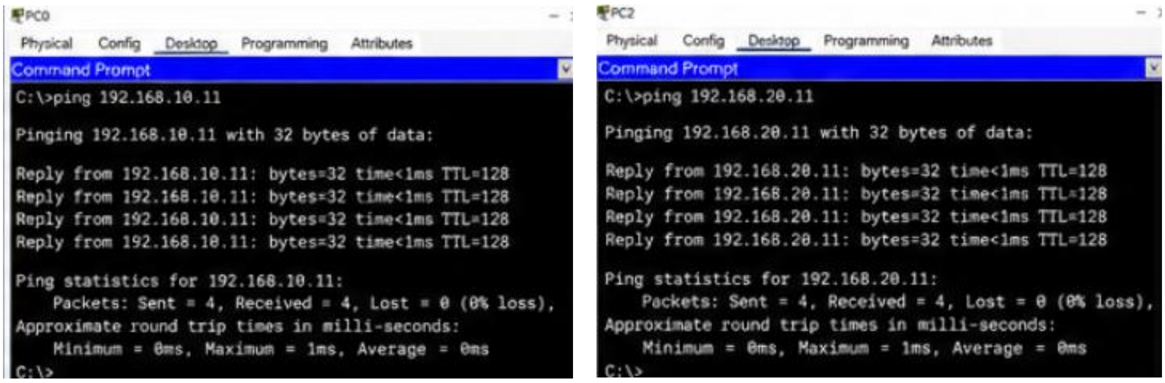


Figure 6: Successful Ping Within the Same VLAN

Step 7: Test Communication Between Different VLANs
From PC0
ping 192.168.20.10
Result
Request timed out
Similarly
From PC1
ping 192.168.20.11
Result
Request timed out
Because there is no router or Layer 3 switch configured, devices in different VLANs cannot communicate.

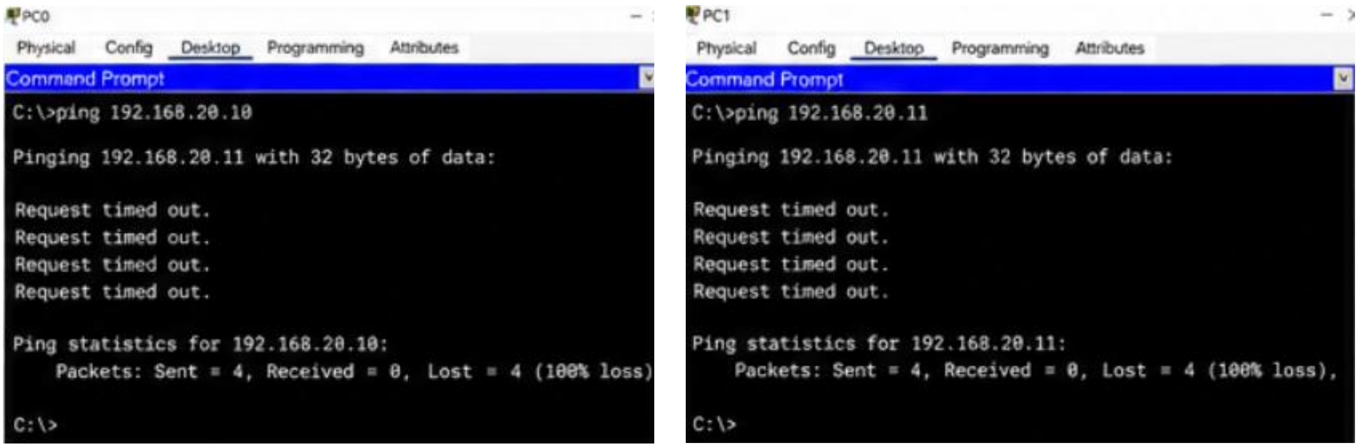


Figure 7: Failed Ping Between Different VLANs

Two VLANs (VLAN 10 and VLAN 20) were successfully created on the switch. Devices within the same VLAN communicated successfully using the ping command, while communication between different VLANs failed because inter-VLAN routing was not configured. This demonstrates how VLANs logically separate network traffic and improve network segmentation.

Q5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

Ans:

Step 1: Create the Network Topology

Place the following devices:

- Router (2911)
- Switch (2960)
- PC0
- PC1
- External Server

Connect them as follows:

- PC0 → Switch
- PC1 → Switch
- Switch → Router (GigabitEthernet0/0)
- Router (GigabitEthernet0/1) → External Server

The router connects the internal LAN to the external network.

Step 2: Configure IP Addresses

Configure the following IP addresses.

Router

Interface	IP Address	Subnet Mask
G0/0 (LAN)	192.168.1.1	255.255.255.0
G0/1 (WAN)	200.1.1.1	255.255.255.0

PC0

Setting	Value
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

PC1

Setting	Value
IP Address	192.168.1.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

External Server

Setting	Value
IP Address	200.1.1.2
Subnet Mask	255.255.255.0
Default Gateway	200.1.1.1

Step 3: Configure Router Interfaces

Open the Router CLI and enter:

```
enable
configure terminal

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 200.1.1.1 255.255.255.0
no shutdown
exit
```

These commands configure the LAN and WAN interfaces.

Step 4: Configure NAT

Enter the following commands:

```
access-list 1 permit 192.168.1.0 0.0.0.255

interface g0/0
ip nat inside
exit

interface g0/1
ip nat outside
exit
```

ip nat inside source list 1 interface g0/1 overload
These commands configure PAT (Port Address Translation), allowing multiple LAN devices to use one public IP address.

Step 5: Verify NAT Configuration
Run the following command:
show ip nat translations
Example Output:

Pro Inside global	Inside local	Outside local	Outside global
icmp 200.1.1.1	192.168.1.10	200.1.1.2	200.1.1.2
icmp 200.1.1.1	192.168.1.11	200.1.1.2	200.1.1.2

The output confirms that the router is translating private IP addresses into the public IP address.

Step 6: Test Connectivity
From **PC0**, open **Desktop → Command Prompt** and execute:
ping 200.1.1.2
Expected output:
Reply from 200.1.1.2:
bytes=32 time<1ms TTL=128

Reply from 200.1.1.2:
bytes=32 time<1ms TTL=128

Reply from 200.1.1.2:
bytes=32 time<1ms TTL=128

Reply from 200.1.1.2:
bytes=32 time<1ms TTL=128
This confirms successful communication with the external server through NAT.

Step 7: Verify Internet Simulation
If the external server is configured with HTTP service:

- Open **Desktop → Web Browser**
- Enter:

http://200.1.1.2
The webpage hosted on the external server should load successfully.

The router was successfully configured with Network Address Translation (NAT). The LAN devices accessed the simulated external server using the router's public IP address. The NAT translation table verified that private IP addresses were translated correctly, demonstrating how NAT enables multiple internal devices to share a single public IP address while accessing external networks.